

Evaluating And Improving Disparity Maps Without Ground Truth

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CCS CONCEPTS

• **Computing methodologies** → **Image processing**: Image segmentation; *Reconstruction*; **Image representations**; **3D imaging**: **Computational photography**.

KEYWORDS

computer vision, stereoscopic 3D, disparity maps

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1 INTRODUCTION

Disparity maps are integral to the fields of computer vision and computer graphics, with many applications in robotics, path planning, computerized drawing, 3D viewing, and more. Stereo algorithms can be used to synthesize disparity maps from stereo image pairs. Unfortunately, most methods to assess disparity map quality rely on a ground-truth disparity map, which is not always available. This paper introduces a method to evaluate disparity map generation algorithms that does not require ground truth, and is instead based on the quality of the disparity maps themselves. This novel scoring metric assesses each pixel in a given disparity map and then provides an overall score for the disparity map. The score is both a visual representation—where each color encodes an error or potential error—and a numerical score. This novel scoring metric is used to compare two top-ranking disparity map generators in different scenarios: Geometry Encoding Volume (IGEV) Stereo [Xu et al. 2023] and Cascaded Recurrent Network (CRE) Stereo [Li et al. 2022]. Our scoring method yields results consistent with intuition and also uncovers, and helps in resolving, an issue with one generator. Namely, the disparity re-scaling done as a final step actually warps the disparities to a range less consistent with ground truth, and this was flagged by our evaluation method. Finally, RANSAC plane-fitting is used to automatically inpaint unknown regions and thereby improve the usability of disparity maps in applications such as non-photorealistic rendering, layer-based stereo methods, and warping.

2 BACKGROUND

There are several existing methods used to assess disparity map quality, the most popular being Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Relative Error (MRE), Bad Matched Pixels (BMP) [Merrouche et al. 2020], Range Multi-Scale Structural Similarity (R-SSIM) [Malpica and Bovik 2009], Sigma Z-Error (SZE) [Cabezas et al. 2011], and BMPRE, which seeks to “combine the advantages of the BMP and the MRE” [Cabezas et al. 2012]. The Middlebury dataset ranks stereo matching algorithms/disparity map generators using the following metrics: the percentage of bad pixels whose error exceeds some threshold (bad 0.5, bad 1.0, bad 2.0, bad 4.0), the average absolute error in pixels (avgerr), root-mean-square-disparity error in pixels (rms), and percent error quantile in pixels (A50, A90, A95, A99) [mid [n. d.]]. All of these methods require ground truth to assess disparity maps, which is not always available. Ground truth disparity maps are usually generated in highly controlled environments using a structured light system, time-of-flight camera, laser rangefinder, or other method. For example, the Middlebury dataset contains stereo image pairs and their disparity maps, generated using structured light [Scharstein and Szeliski 2003]. This is not attainable for the average photographer taking homemade stereo photos using a 3D camera.

3 APPROACH

In this paper, we assess the quality of a disparity map without needing to use ground truth. We do so by leveraging several known facts about a good disparity map. A good disparity map must fulfill certain requirements. The majority of the pixels need to fuse. If the disparity of pixel (x, y) in the left disparity map is d such that the corresponding pixel on the right is given by $(x - d, y)$, then the two pixels ought to match, within some small tolerance, or the disparity map is wrong. By ‘match’, we mean that the two pixels need to map to each other and be similar in color. There should be few unknown pixels/regions. These are usually depicted in black, as is the case with Middlebury disparity maps. Edges should be smooth and crisp for optimal viewing. This is the most important aspect of fusion [Istead et al. 2021]. The disparity map should fill in the leftmost vertical strip in the left image and the rightmost vertical strip in the right image properly. This is difficult because there is no matching to do for these regions; they are endemic to their own left or right image and do not appear in the corresponding right or left image, respectively – but they still need to be filled in. Many algorithms streak and blend these regions heavily in disparity maps, at the expense of the viewing experience.

Our method also corrects certain erroneous pixels. It enables users to compare stereo methods and improve their outputs, and the higher quality thereof will be beneficial to the application using the disparity maps.

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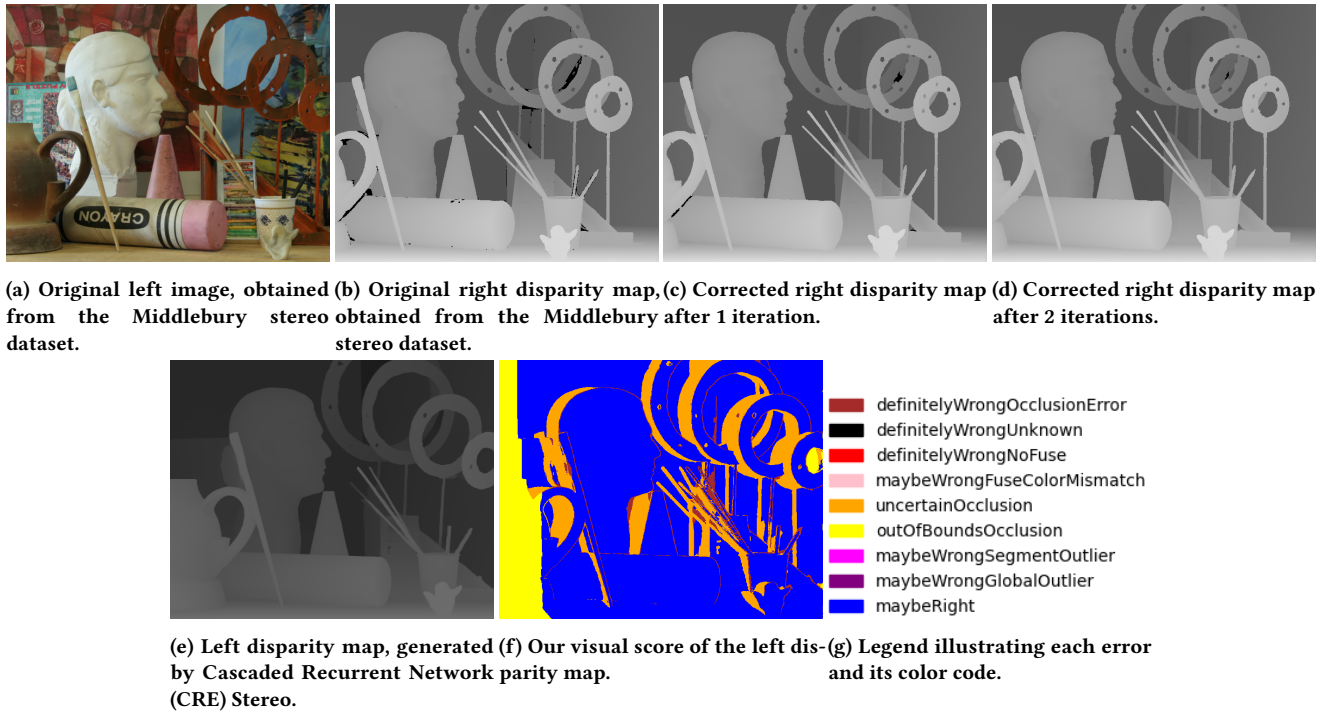


Figure 1: Our method identifies various errors in disparity maps and can be applied iteratively to improve a disparity map.

The following is a high-level overview of our scoring metric. We first identify definitively wrong disparities if they result in obvious errors. Pixels in unknown regions of the disparity map are black, and always wrong. Pixels may encode occlusion in the wrong direction. Pixels may not fuse properly. We then segment the image based on both edges and color. We use a Simple Linear Iterative Clustering segmentation algorithm, which segments an image using k-means clustering, to do this. We extract all the pixels from the image and disparity map that belong to a given segment. We consider all the disparities associated with that segment, keeping in mind that disparities should be continuous. We identify all the discrete, consecutive/contiguous disparity groups within the segment. If any group stands out from the rest, we mark those disparities as potential outliers. Next, we repeat the previous step on the entire disparity map to identify potential global outliers. We assign each pixel a color-coded score and output an image based on the disparity map where the color of each pixel reflects the likelihood that the pixel is wrong. Finally, we use RANSAC plane-fitting to correct outliers and unknown pixels.

4 RESULTS

We used our method to obtain several findings. Firstly, we found that Iterative IGEV outperforms CRE on our batch of homemade stereo image pairs. Also, CRE’s performance was found to vary more drastically in terms of disparity map quality; some disparity maps earned a nearly perfect score, while others were shockingly bad. IGEV’s performance was more consistent. Furthermore, we compared the performance of IGEV as evaluated on homemade

stereo images vs Middlebury’s test dataset. Error scores are, unsurprisingly, lower when the algorithm is evaluated on Middlebury stereo images. Homemade stereo images are more error-prone because of the lack of tools and setup involved in their generation.

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